

Edge AI Computing for Grape Leaf Disease Detection



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Thesis Progress...

February 8, 2026

Proposed System Architecture

- *Four-Stage Pipeline:*

- Stage 1: Input image captured using ESP32-S3 camera
- Stage 2: Localization: A lightweight model to detect leaf region (YOLO)
- Stage 3: Classification: Disease detection model (MobileNetV2)
- Stage 4: Sending only the classification result via LoRaWAN

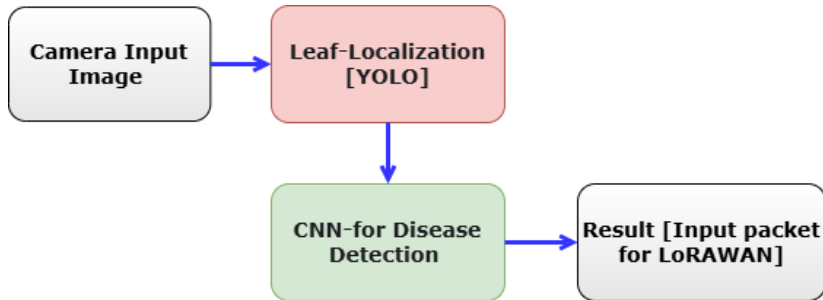


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Introduction

- This work is focused on detecting grape leaf disease by deploying Deep Learning models at the 'edge' using the ESP32-S3. I'm developing a two-stage on-device pipeline: **first**, localization model isolates the leaf from the camera input image, and **second**, fine-tuned MobileNet model that classif disease.
- Today, I will present my progress on the classification model, the challenges of real-world data, and my road map for the final hardware integration.

Leaf Disease Classifier Model Training

- In model selection process I benchmarked two popular architectures: **ResNet** and **MobileNetV2**. While both gave me excellent accuracy, **ResNet** was simply too heavy for ESP32. So, **MobileNetV2** was chosen due to its lightweight architecture and ensuring compatibility with the **ESP32-S3** resource constraints.

Dataset Overview: Grape Leaf Disease

- **Source:** Publicly available dataset (Kaggle), contains 4 distinct class of diseases.
- **Data Distribution:** The dataset is **balanced**, with approx. 2,100 to 2,400 images per class, preventing class-imbalance bias during training.
- **Split Strategy:** Standard **80/20 Split** (80% Training, 20% Testing).

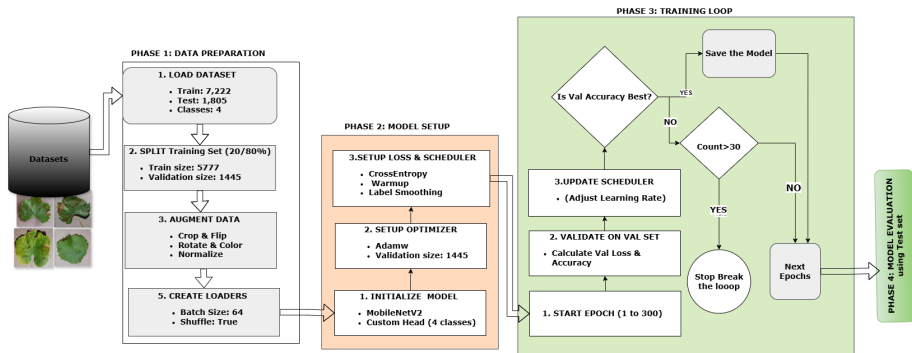
Disease Class	Training (80%)	Testing (20%)	Total Images
Black Rot	1,888	472	2,360
Esca (Black Measles)	1,920	480	2,400
Healthy	1,692	423	2,115
Leaf Blight (Isariopsis)	1,722	430	2,152
Total	7,222	1,805	9,027

Table: Distribution of images across disease classes.

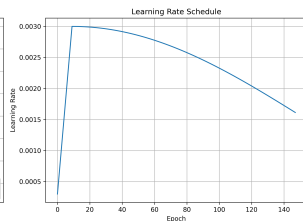
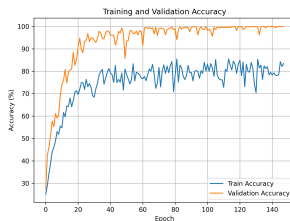
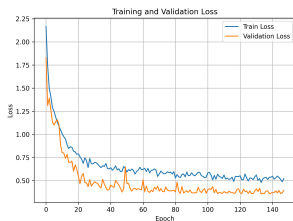
Source: <https://www.kaggle.com/datasets/bmshahriaalam/grape-leaf-disease>

Leaf Disease Classifier: MobileNetV2 Model Training Pipeline

- This diagram illustrates the training workflow I implemented. I want to highlight two components: **Data Augmentation** and **Scheduler**

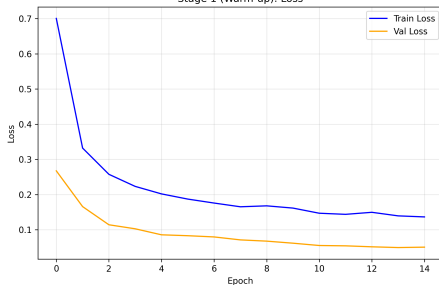


Learning Curve of MobileNetV2

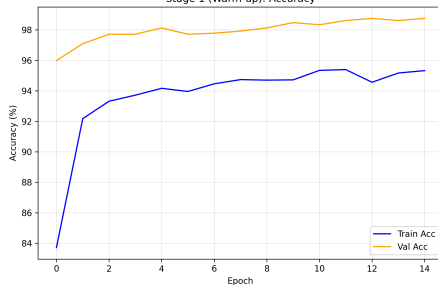


MobileNetV2 Transfer Learning and Fine-tuning Curves

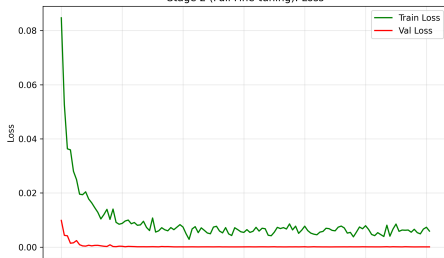
Stage 1 (Warm-up): Loss



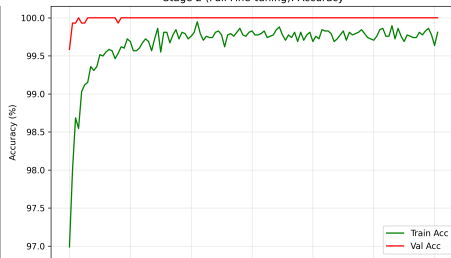
Stage 1 (Warm-up): Accuracy



Stage 2 (Full Fine-tuning): Loss



Stage 2 (Full Fine-tuning): Accuracy



Leaf Disease Classifier Model Predictions

True: Esca
Pred: Esca



True: Black_rot
Pred: Black_rot



True: Esca
Pred: Esca



True: Esca
Pred: Esca



True: Esca
Pred: Esca



True: Healthy
Pred: Healthy



True: Leaf_blight
Pred: Leaf_blight



True: Leaf_blight
Pred: Leaf_blight



Critical Challenge: Dataset vs. Real-World Input

● Input Discrepancy:

- *Training*: Single, centered leaves (Uniform background).
- *Inference*: Complex scenes (Multiple leaves, branches, soil, and sky).

● The Consequence:

- Direct classification fails because the model analyzes background noise instead of the leaf texture.

● The Gap (Data Scarcity):

- We need **YOLO** to localize/crop the leaf.
- **Problem**: The public datasets from kaggle have *class labels* (classification) but lack *bounding boxes* (detection).

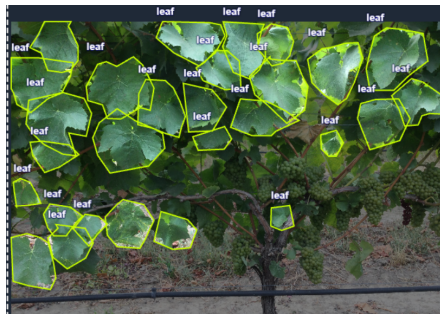
Solution Strategy:

- 1 Create a custom dataset for Localization
- 2 Annotate bounding boxes for "Leaf" vs "Background"
- 3 Train lightweight YOLO to crop ROI

Data Preparation for Leaf Localization (YOLO)



(a) Raw Camera Input



(b) Ground Truth Annotation

- **The Requirement:** To train the YOLO model, the system requires "Ground Truth" data where every leaf instance is explicitly defined.
- **Annotation Process:**
 - Manually labeling specific "Leaf" regions in complex vineyard scenes.
 - Creating a custom dataset that pairs images with **Spatial Coordinates** (Bounding Boxes/Polygons).

ESP32 Development Kit: Freenove ESP32-S3

Selected Device Specifications:

- **SoC:** Espressif ESP32-S3 (Dual-Core Xtensa LX7, 240 MHz).
- **AI Capabilities:** Built-in **Vector Instructions** for Neural Network acceleration.
- **PSRAM:** **8MB** , Memory Storage: **16MB**, **1GB** memory card Can also be inserted.

Connectivity & Interface:

- **Camera:** OV2640 Support (Vision Input).
- **Comms:** UART (for LoRaWAN Node integration).



Freenove ESP32-S3 WROOM

Conclusion & Next Steps

Current Achievements:

- **Model Validation:** Successfully fine-tuned **MobileNetV2** with high accuracy.
- **Hardware Procurement:** Selected and ordered **Freenove ESP32-S3 (8MB PSRAM)** as a target edge device.

Next Steps:

- **Dataset Engineering:** Complete the manual annotation, Or Create an augmented dataset for YOLO.
- **YOLO Training:** Train a lightweight detection model to automate ROI extraction.
- **Integration:** Deploy both models onto the ESP32-S3 and establish UART communication with the Nucleo-WL55JC for LoRaWAN transmission.

References



Xu Tao, Jackson Butcher, Simone Silvestri, and Sajal K. Das (2025)
iCrop+: An Edge-boosted Crop Disease Detection System via TinyML and LoRa Communication.

In 2025 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops), pp. 558–560.



Md Jawadul Karim, Md Omaer Faruq Goni, Md Nahiduzzaman, Mominul Ahsan, Julfikar Haider, and Marcin Kowalski (2024)

Enhancing agriculture through real-time grape leaf disease classification via an edge device with a lightweight CNN architecture and Grad-CAM.

Scientific Reports, 14(1), 16022.



Ning Zhang, Enxu Zhang, Guowei Qi, Fei Li, and Cheng Lv (2025)

Lightweight grape leaf disease recognition method based on transformer framework.

Scientific Reports, 15(1), 28974.



Grape Leaf Disease Dataset:

<https://www.kaggle.com/datasets/bmshahriaalam/grape-leaf-disease>

Thank You

Questions?